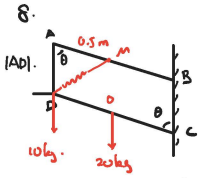
8. ($|AD|=0.5m$)



不用力一题通:

$E_p = |OC|\cdot 2mg\cdot\cos\theta + |CO|\cdot mg\cdot\cos\theta + \frac{1}{2}k\cdot|DM|^2$

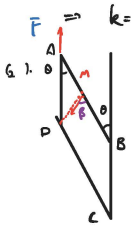
$m=10kg$.

$|DM|^2 = |AD|^2+|AM|^2-2\cos\theta\cdot|AD|\cdot|AM|$

$E_p = \cos\theta\cdot[(2|OC|+|CO|)mg - k\cdot|AD|\cdot|AM|] + C$ — 常数

$\Rightarrow \frac{dE_p}{d\cos\theta}=0$

$\Rightarrow k = \frac{2|CO|}{|AD|\cdot|AM|}mg$



AD 轻杆. 受力沿杆

对 AB 杆. 以B为支. $\Sigma M=0$

$F\cdot|AB|\cdot\sin\theta = |DM|\cdot k\cdot|BM|\cdot\sin\beta$

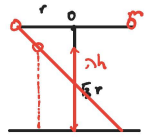
正弦 $\frac{\sin\theta}{|DM|}=\frac{\sin\beta}{|AD|}$

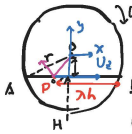
$F = \frac{|DM|\,k}{|AB|\sin\theta}\cdot|BM|\cdot\frac{\sin\beta}{|AD|}\cdot|AD|$

$= \frac{|BM|\cdot|AD|}{|AB|}\cdot\frac{2|CO|}{|AD|\cdot|AM|}mg$

$= mg\cdot\frac{2|BM|}{|AM|}$

2.





小球高 λh 时. 盘 ω_λ=?

高 $z=\lambda h$.

$|AB|_{水平}=\sqrt{3}r$. $|AB|_{竖}=h=\sqrt{3}\lambda r$

小球在 P 点. $|BP|_{水平}=\lambda h$.

记 $U_2=-\dot z$. 在 ω_λ 系中. 小球速度

竖: U_2 向下. 水平向右 U_B.

地系: $\vec U_P = -\hat z U_2 + \hat A_B U_2 + \vec\omega_\lambda\times\vec{OP}$ ($|HP|_{水平}$ ↑)

$= -\hat z U_2 + \hat x(U_2 - \omega_\lambda\frac{r}{2}) + \hat y\cdot(\omega_\lambda\cdot(\lambda h-\frac{h}{2}))$ ($\frac{h}{2}=(\lambda-\frac{1}{2})\sqrt{3}r$)

角动量守恒.

$L_z = m\cdot U_2\cdot\frac{r}{2} - m\cdot\omega_\lambda[(\frac{r}{2})^2+(\lambda-\frac{1}{2})^2 3r^2]$

$-2\cdot mr^2\cdot\omega_\lambda = 0$ ①

能量守恒.

$\frac{1}{2}\cdot 2mr^2\omega_\lambda^2 + \frac{1}{2}mU_2^2 + \frac{1}{2}m\cdot(U_2-\omega_\lambda\frac{r}{2})^2$ ②

$+\frac{1}{2}m\cdot(\omega_\lambda\cdot(\lambda-\frac{1}{2})\sqrt{3}r)^2 = mg\sqrt{3}r(1-\lambda)$

①: $U_2 = 2\cdot\omega_\lambda\cdot r\cdot[\frac{1}{4}+3(\lambda-\frac{1}{2})^2+2]$. 代入②.

$\frac{1}{2}mr^2\omega_\lambda^2\cdot[2+2\cdot[\frac{9}{2}+6(\lambda-\frac{1}{2})^2]^2 - 2\cdot\frac{1}{2}\cdot(\frac{9}{2}+6(\lambda-\frac{1}{2})^2)$

$+\frac{1}{4}+3(\lambda-\frac{1}{2})^2] = mg\sqrt{3}r(1-\lambda)$

$\omega_\lambda = \sqrt{\frac{2}{r}}\cdot\sqrt{\frac{\sqrt{3}(1-\lambda)}{9/4+3(\lambda-\frac{1}{2})^2-(\frac{9}{2}+6(\lambda-\frac{1}{2})^2)+2\cdot(\frac{9}{2}+6(\lambda-\frac{1}{2})^2)^2}}$

---

(3) 只要①式.

$U_2 = 2\cdot\omega_\lambda\cdot r\cdot[\frac{1}{4}+3(\lambda-\frac{1}{2})^2+2]$

$\Rightarrow -\sqrt{3}\dot\lambda = 2\omega_\lambda\cdot[\frac{9}{4}+3(\lambda-\frac{1}{2})^2]\cdot$ ($z=\sqrt{3}r\lambda$, $U_2=-\sqrt{3}r\dot\lambda$)

$\Rightarrow\int_0^1\frac{\sqrt{3}\cdot d\lambda}{\frac{9}{2}+6(\lambda-\frac{1}{2})^2} = \int_0^T\omega_\lambda\,dt$

$\int_0^1\frac{\sqrt{3}\cdot d\lambda}{\frac{9}{2}+6(\lambda-\frac{1}{2})^2} = \frac{\sqrt{3}}{6}\cdot\int_0^1\frac{d\lambda}{(\lambda-\frac{1}{2})^2+\frac{3}{4}}$

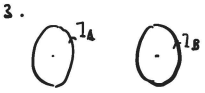
$\lambda-\frac{1}{2} = \frac{\sqrt{3}}{2}\cdot\tan\theta$ $\tan\theta=-\frac{\pi}{6},\frac{\pi}{6}$

$= \frac{\sqrt{3}}{6}\int\frac{\frac{\sqrt{3}}{2}\,d\tan\theta}{\frac{3}{4}\cdot(\tan^2\theta+1)} = \frac{\sqrt{3}}{6}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\cdot\int_{-\frac{\pi}{6}}^{+\frac{\pi}{6}}\frac{\frac{1}{\cos^2\theta}\,d\theta}{\frac{1}{\cos^2\theta}}$

$= \frac{1}{3}\cdot\frac{\pi}{3}\cdot$

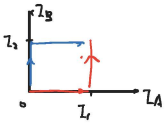
$\Rightarrow\int_0^T\omega_\lambda dt = \frac{\pi}{9}$

3.



$\Phi_A = L_A I_A + M_{AB} I_B$

$\Phi_B = L_B I_B + M_{BA} I_A$. $I_A - I_B = 0$

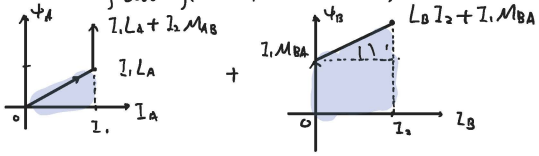


$dW = dq\cdot U$

$= \frac{dq}{dt}\cdot U\,dt$

$= I\,d\Psi$

$\int dW = \int(I_A\,d\Psi_A + I_B\,d\Psi_B)$.



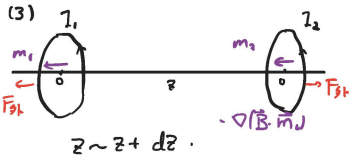
$W = \frac{1}{2}I_1^2L_A + I_2\cdot I_1M_{BA} + \frac{1}{2}L_BI_2^2$

交换顺序:

$W' = \frac{1}{2}I_1^2L_A + I_1\cdot I_2M_{AB} + \frac{1}{2}L_BI_2^2$

$W=W' \Rightarrow M_{AB}=M_{BA}$

(3)



$\Phi_A = L_AI_A + M_{AB}I_B$

×

$z\sim z+dz$.

$dW_{电之} = F_{引}\,dz + I_A\,d\Psi_A + I_B\,d\Psi_B$

$I_1I_2\,dM_{AB} = F_{引}\,dz + dM_{AB}\,I_1I_2 + dM_{BA}\,I_1I_2$

$\Rightarrow F_{引}\,dz = -dM_{AB}\cdot I_1I_2$

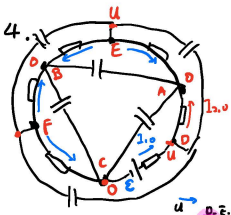
$M_{AB} = \cdot\frac{2\cdot\mu_0\cdot\pi r_1^2 I_2}{4\pi\cdot z^2}\cdot\frac{1}{I_2}\cdot\pi r_2^2 = \frac{\pi}{2}\frac{r_1^2r_2^2\mu_0}{z^3}$

I_1 在 2. 产生 B

$\Rightarrow F_{引} = I_1I_2\cdot\frac{3\pi}{2}\frac{r_1^2r_2^2\mu_0}{z^4}$

相互作用吸引. 大小等 F_B

4.



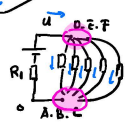
(1) $t\to 0$. $q_C=0$.

电容两端 $\Delta U=0$

$U=\mathcal{E}-I_{10}R_1$

$I_{20}R_2=U$.

$\frac{\mathcal{E}-U}{R_1}=\frac{U}{R_2}+\frac{U}{R_3}+\cdots+\frac{U}{R_6}$.



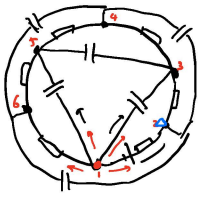
$\Rightarrow U_0=\frac{\mathcal{E}/R_1}{\frac{1}{R_1}}$

$\Rightarrow I_{10}=\frac{\mathcal{E}}{R_1+(\sum_{i=2}^{6}R_i^{-1})^{-1}}$; $I_{20}=\frac{\frac{\mathcal{E}}{R_1}\frac{1}{R_2}}{\sum_{i=1}^{6}\frac{1}{R_i}}$

(2) $t\to\infty$ 电容电流 $\to 0$ $\Rightarrow$ 电压表.

$I_{10}=I_{20}=\frac{\mathcal{E}}{\sum_{i=1}^{6}R_i}$

(3).



电势 $U_1, U_2\cdots U_6$. 电阻一样.

$\frac{U_1-U_2}{r}+\frac{U_1-U_6}{r}+C[\dot U_1-\dot U_3]+C(\dot U_1-\dot U_5)=-\frac{\mathcal{E}}{r}$ ↙ △

$\frac{U_2-U_3}{r}+\frac{U_2-U_1}{r}+C(\dot U_2-\dot U_4)+C(\dot U_2-\dot U_6)=+\frac{\mathcal{E}}{r}$ ↙ △

其它 $\to 0$

$U_i=U_{i+6}$ $U_7=U_1$ $U_8=U_2$.

除 1.2 之外: $\frac{U_n-U_{n+1}}{r}+\frac{U_n-U_{n-1}}{r}$

$+C(\dot U_n-\dot U_{n+2})+C(\dot U_n-\dot U_{n-2})=0$

猜解. $U_n=A\,e^{\frac{\lambda}{rC}t}+\alpha\cdot n$. $\alpha\in C$.

$\Delta e^{i\frac{\pi}{3}}$

$\Rightarrow 2-e^{\alpha}-e^{-\alpha}+\lambda[2-e^{2\alpha}-e^{-2\alpha}]=0$.

$\lambda=-\frac{2-e^{\alpha}-e^{-\alpha}}{2-e^{2\alpha}-e^{-2\alpha}}$. $U_n=U_{n+6}$.

$e^{6\alpha}=1$. $\alpha=mi\frac{2\pi}{6}$ $m=0,1,2\cdots$ $\frac{1-\cos 60^\circ}{1-\cos 120^\circ}$

$m=0$. U 常数.

$m=1.5$. $\lambda_{1,5}=-\frac{2-2\cos\alpha}{2-2\cos 2\alpha}=-\frac{1}{3}$

$m=2,4$. $\lambda_{2,4}=-\frac{1-\cos 60^\circ}{1-\cos 120^\circ}=-1$. $\frac{1-\frac{1}{2}}{1+\frac{1}{2}}$

$m=3$. $\lambda_3=-\frac{1}{2}$. $-\frac{1-\cos 90^\circ}{1-\cos 180^\circ}$

⇒ 取实根:

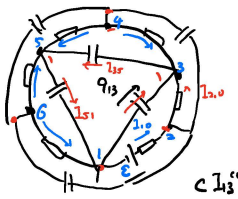
$U_n=e^{-\frac{1}{2}\frac{t}{rC}}\cdot A\cdot(-1)^n$ ↙ $m=3$

$+e^{-\frac{t}{3rC}}\cdot B\cdot\cos(n\frac{2\pi}{6}+\varphi_B)$ $m=1.5$

$+e^{-\frac{t}{rC}}\cdot C\cdot\cos(n\frac{4\pi}{6}+\varphi_C)$ $m=2,4$

代入 $t=0$. 定系数 + $\frac{\mathcal{E}}{r}$ 特解. 猜 $U=A+Be^{-\frac{t}{rC}}$ $t\to\infty$ 代入

简化版:



求 $\dot I_1(0)=?$ $\dot I_2(0)=?$

C-D-A.

$U_1-U_3=I_1R_1+I_2R_2-\mathcal{E}$

↓求导

$C\dot I_{13}^{(0)}=\dot U_1-\dot U_3=\dot I_1R_1+\dot I_2R_2$

$\dot U_1-\dot U_3=\frac{\dot q_{13}}{C}$

令 $U_0=\frac{\mathcal{E}/R_1}{\sum_{n=1}^{6}\frac{1}{R_n}}$

$\begin{cases}I_6+I_{51}=I_1+I_{13}\\ I_2+I_{13}=I_3+I_{35}\\ I_4+I_{35}=I_5+I_{51}\end{cases}$; $I_{13}+I_{35}+I_{51}=0$

$I_1=U_0\frac{\mathcal{E}}{\sum_{n=2}^{6}\frac{1}{R_n}}\cdot\frac{1}{R_1}$. $I_2=U_0\frac{1}{R_2}$; $I_3=-U_0\frac{1}{R_3}$.

$I_4=U_0\frac{1}{R_4}$. $I_5=-U_0\frac{1}{R_5}$. $I_6=\frac{U_0}{R_6}$.

$I_6-I_{35}-I_{13}=I_1+I_{13}$

$I_2+I_{13}-I_3=I_{35}$

$I_6-I_2-I_{13}+I_3-I_{13}=I_1+I_{13}$

$\Rightarrow I_{13}=\frac{I_6-I_2+I_3-I_1}{3}$;

同理: $I_{24}=\frac{I_1-I_3+I_4-I_2}{3}$

$I_{35}=\frac{I_2-I_4+I_5-I_3}{3}$

⋮

$\begin{cases}\dot I_1+\dot I_2=rC\cdot I_{13}\\ \dot I_2+\dot I_3=rC\,I_{24}\\ \dot I_3+\dot I_4=rC\,I_{35}\\ \dot I_4+\dot I_5=rC\,I_{46}\\ \dot I_5+\dot I_6=rC\,I_{51}\end{cases}$ $\times[\dot I_6+\dot I_1=rC\,I_{62}]$.

$\dot I_1+\dot I_2+\cdots+\dot I_6=0$

$\Rightarrow \dot I_1\sim$. $\dot I_2=\checkmark$

$\dot I_3=rC\dot I_{51}-\dot I_6$

$\dot I_4=rC\dot I_{46}-rC\dot I_{51}+\dot I_6$

$\dot I_5=rC(\dot I_{35}-\dot I_{46}+\dot I_{51})+\dot I_6$

$\dot I_6=rC(\dot I_{24}-\dot I_{35}+\dot I_{46}-\dot I_{51})-\dot I_6$

$\dot I_1+\dot I_6+rC\cdot(\dot I_{24}+\dot I_{46})=0$

$\dot I_1-\dot I_6+rC\cdot(\dot I_{24}-\dot I_{35}+\dot I_{46}-\dot I_{51})=rC\dot I_{13}$

$\dot I_1=(\frac{\dot I_{51}}{2}+\dot I_{13}+\dot I_{24}-\dot I_{46})\,r$

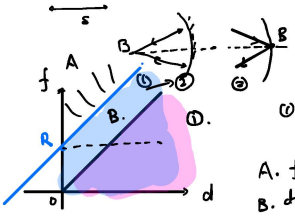
5.



一 B 成像在 B 点.

对凹面

$\frac{1}{s}+\frac{1}{d}=\frac{1}{f}$ ①



$\frac{1}{s}+\frac{1}{d+R}=\frac{1}{f}$ ②

$d>0$

① $d>f$. ② $d+R>f$. $d>0$.

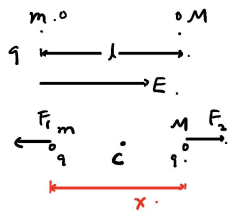
A. $f>d+R$. 无解. $d>0$ $d>f-R$

B. $d+R>f>d$. $d>0$. ②: $s=\frac{(d+R)f}{(d+R)-f}$

C. $d>f>0$. ① $s=\frac{df}{d-f}$

② $s=$ 负

6.



(1) $U_{0,}=U_{0}=0$

质心加速度 $a_c=\frac{2Eq}{m+M}$.

以 a_c 为系：外力

$F_1=-Eq+m\cdot\frac{2Eq}{m+M}$

$=\frac{m-M}{m+M}\cdot Eq$ ＋

$F_2=Eq-M\frac{2Eq}{m+M}$

$=\frac{m-M}{m+M}Eq$. －

质心 C 不动. $F_1\sim F_2$ → 有效势能 $-x\cdot F_1$

$\Delta E=0$: $\frac{1}{2}\cdot\mu\dot{x}^2-x\cdot F_1+\frac{Kq^2}{x}=-lF_1+\frac{Kq^2}{l}$

最远 $\dot{x}=0$, $x=x_m$

$\Rightarrow -x_mF_1+\frac{Kq^2}{x_m}=-lF_1+\frac{Kq^2}{l}$

$(x_m-l)F_1=-\frac{Kq^2(x_m-l)}{x_ml}$

$\Rightarrow x_{m_1}=l$. x_{m_2}: $Eq\frac{m-M}{m+M}=-\frac{Kq^2}{x_ml}$

$m<M$ 时 $x_{m_2}>0$

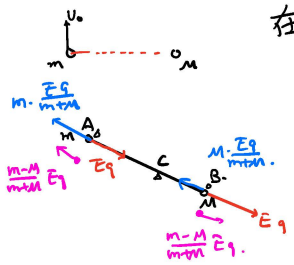
$x_m=-\frac{Kq(m+M)}{lE\cdot(m-M)}$.

1). $\frac{-Kq(m+M)}{lE\cdot(m-M)}>l$. $x_{max}=\frac{-Kq(m+M)}{lE\cdot(m-M)}$

2) $\frac{-Kq(m+M)}{lE\cdot(m-M)}<l$. $x_{max}=l$.

(2). $M=2m$. $mU_0^2=\frac{Kq^2}{l}$. $E=\frac{Kq}{4l^2}$

由动 $m\to M$ 始终成立.



在质心系 a_c 中. 看见两体问题.

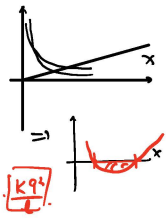
相互作用力 $F=\frac{m-M}{m+M}Eq+\frac{Kq^2}{x^2}$

$\to E_p=-x\cdot\frac{m-M}{m+M}Eq+\frac{Kq^2}{x}$

$=\frac{Kq^2}{l}\cdot\left[\frac{x}{l}\cdot(+\frac{1}{3})\cdot\frac{1}{4}+\frac{l}{x}\right]$

$E_k=\frac{1}{2}\frac{m\cdot M}{m+M}(\dot{x}^2+x^2\dot{\theta}^2)$

$=\frac{1}{2}\cdot\frac{2}{3}m(\dot{x}^2+x^2\dot{\theta}^2)$



$\Delta L=0$. $\mu\cdot x^2\dot{\theta}=\mu\cdot l\cdot U_0$

$\Rightarrow$

$\Delta E=0$: $\frac{1}{2}\cdot\frac{2}{3}m\dot{x}^2+\frac{1}{2}\cdot\frac{2}{3}m\cdot(\frac{x}{l})^2\cdot l^2\cdot\frac{l^2U_0^2}{x^4}$ $\frac{Kq^2}{l}$

$+\frac{Kq^2}{l}\left[\frac{x}{l}\cdot(+\frac{1}{3})\cdot\frac{1}{4}+\frac{l}{x}\right]$

$=\frac{1}{2}\cdot\frac{2}{3}mU_0^2+\frac{Kq^2}{l}\cdot[+\frac{1}{12}\cdot+1]$.

远 $\dot{x}=0$. 令 $\frac{l}{x}=\xi$ $\xi=1\sim$

$\frac{1}{3}\cdot\xi^2+(+\frac{1}{12}\xi+\xi)=\frac{1}{3}+(+\frac{1}{12}+1)$

$\frac{1}{3}(\xi-1)(\xi+1)-\frac{1}{12}\frac{\xi-1}{\xi}+(\xi-1)=0$

$\frac{1}{3}(\xi-1)(\xi+1)-\frac{1}{12}\frac{\xi-1}{\xi}+(\xi-1)=0$

$\xi\neq1$. $\frac{1}{3}(\xi+1)-\frac{1}{12\xi}+1=0$

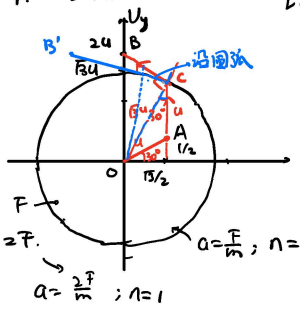
$4\xi^2+4\xi-1+12\xi=0$ $\xi_+\xi_-=-1$

$\xi=-2\pm\frac{\sqrt{17}}{2}$ 取正. $\xi_+=\frac{1}{-2+\frac{\sqrt{17}}{2}}$.

$\to\frac{1}{\xi}\infty$. $x_m=\frac{l}{\xi_+}=2l\cdot(4+\sqrt{17})$.

---

7. 弦 $u/2$



以风为系

$dt=\frac{du}{a}\sim$

$du\sim dx$.

$\vec{v}\sim\vec{x}$

$\frac{1}{v}\sim n$

$dt\sim ds$. ← 光程

等效光学 内 $n=2$.

外 $n=1$

求光程最小

$A\to C\to B$ 始终满足折射定律.

$AC\perp x$ 轴.

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8. m_0 $E=\frac{m_0c^2}{\sqrt{1-v^2/c^2}}$ $1-\frac{v^2}{c^2}=\frac{m_0c^2}{E}$

$1-\frac{m_0c^2}{E}=\frac{v^2}{c^2}$ $\frac{v}{c}\to1$. $\frac{v}{c}=1-\varepsilon$

$\Rightarrow 1-\frac{m_0c^2}{E}=1-2\varepsilon$ $(+\varepsilon^2)$

$\varepsilon\simeq\frac{m_0c^2}{2E}$

$\Delta t=\frac{l}{v_1}-\frac{l}{v_2}=\frac{l}{c}(1+\varepsilon_1)-\frac{l}{c}(1+\varepsilon_2)$

$=\frac{l}{c}(\varepsilon_1-\varepsilon_2)=\frac{l}{c}\cdot(\frac{m_0c^2}{2\cdot0.1MeV}-\frac{m_0c^2}{2\cdot100MeV})$

$\sim m_0c^2\sim\frac{\Delta t\cdot\cdot2\cdot0.1MeV}{l/c}$ ($\Delta t\sim0.3s$; $l/c\hookrightarrow$ 75亿年)

$\sim\frac{1\times10^5\ eV}{75\times10^8\times365\times24\times60\times60}$